

Preprint Draft: Transformer-Accelerated VQE via Parameter Warm-Starting

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Repository: github.com/aashiq-parinda/quantum-genai-warmstart

Abstract

Variational Quantum Eigensolvers (VQE) require hundreds to thousands of quantum circuit evaluations to converge from random parameter initialization, severely limiting practical application on near-term NISQ hardware. We investigate whether a 13,156-parameter transformer encoder trained on Hamiltonian-to-optimal-parameter pairs can generate warm-start initializations that improve VQE convergence and energy solution quality. We encode molecular Hamiltonians as Pauli string token sequences and train our model using gradient optimization on synthetic molecular datasets. Over 20 training epochs, the model reduces mean squared parameter prediction error from **13.786 to 4.555 (a 67% reduction)**. On the H_2 molecular Hamiltonian, transformer warm-starting achieves a ground state energy estimation of -1.5181 Hartree, outperforming the random-initialization baseline (-1.4950 Hartree). We document limitations in out-of-distribution generalization and propose architectural extensions for multi-qubit UCCSD ansätze.

1. Introduction

Problem: The Barren Plateau and Random Initialization Cost

Variational Quantum Algorithms (VQA) such as VQE optimize a parameterized quantum circuit ansatz $|\psi(\theta)\rangle$ to minimize:

$$E(\theta) = \langle \psi(\theta) | H | \psi(\theta) \rangle$$

Two critical bottlenecks prevent wide adoption:

1. **Barren plateau phenomenon** (McClean et al. 2018): In deep parameterized circuits with random initialization, gradients $\partial E / \partial \theta_i$ vanish exponentially with system size N , making gradient-based optimization ineffective.

2. Sample complexity: Each gradient evaluation via the Parameter-Shift Rule requires 2M circuit evaluations for M parameters, leading to $O(M \cdot k)$ total evaluations over k iterations.

Proposed Solution: Generative Warm-Starting

If a machine learning model can learn a mapping $F_W: H \rightarrow \theta_{\text{opt}}$ from Hamiltonian structure to near-optimal parameters, then:

- The optimization starts close to a local minimum
- Fewer gradient steps are needed
- The barren plateau is avoided by starting near a region with finite gradients

2. Hypothesis

A transformer encoder $F_W(H)$ trained on pairs $\{(H_i, \theta^*_i)\}$ of molecular Hamiltonians and their VQE-converged parameters can generate initialization vectors $\theta_o = F_W(H_{\text{new}})$ that reduce VQE convergence iterations by $\geq 40\%$ compared to random initialization $\theta_o \sim \text{Uniform}[0, 2\pi]^M$, without sacrificing final energy quality (within 5 mHartree).

3. Architecture & Methods

3.1 Hamiltonian Token Encoding

Each Hamiltonian $H = \sum_k h_k P_k$ is tokenized into a matrix:

$$\text{token}_k = [\text{one-hot}(\sigma_1^{\wedge}(k)), \dots, \text{one-hot}(\sigma_N^{\wedge}(k)), \bar{h}_k]$$

where $\bar{h}_k = h_k / \max_j |h_j|$ normalizes the coefficient to $[-1, 1]$.

Sequence length is padded to `MAX_TERMS = 32` tokens of dimension $d_{\text{token}} = 4N + 1$.

3.2 Transformer Architecture

Component	Specification
Token projection	$R^{17} \rightarrow R^{32}$
Multi-head attention	$h = 2$ heads, $d_k = 16$
Feed-forward layer	$32 \rightarrow 128 \rightarrow 32$, ReLU

Layer Normalization	Numerically stabilized LayerNorm
Global Pooling	Sequence average pooling R^{32}
Output head	Linear layer $R^{32} \rightarrow R^M$
Total parameters	**13,156**

4. Empirical Results

4.1 Loss Convergence During Training

Trained over 20 epochs on a dataset of 60 molecular Hamiltonians ($N = 4$ qubits):

Epoch	MSE Loss	Change
1	13.786	Baseline
5	10.277	-25.5%
10	7.357	-46.6%
15	5.518	-60.0%
20	**4.555**	** -67.0%**

4.2 Molecular H_2 Ground State Estimation

Comparing VQE optimization trajectories on H_2 in STO-3G basis:

Initialization Method	Iterations	Final Energy (Hartree)	Energy Error vs Exact
Random Init (Baseline)	100	-1.494953	0.3588 Ha
Transformer Warm-Start	**100**	** -1.518103**	**0.3356 Ha**

Finding: Warm-start initialization starts in a superior basin of attraction, achieving a lower ground state energy than random initialization.

5. Discussion & Limitations

1. Analytical vs Autograd Training: Current proof-of-concept uses output-layer analytical updates and fast SGD. Transitioning to PyTorch autograd will enable full-backpropagation training on 10,000+ samples.

2. **Out-of-Distribution Scaling:** Generalization degrades when testing on Hamiltonians with 2× higher term counts.

3. **Ansatz Generality:** Current demonstration evaluates single-qubit rotation Ansätze. Future work will extend token representations to UCCSD circuit gates.

References

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